DISCRETE MATH · COUNTING AND PROBABILITY

Counting Outcomes Explorer

Explore & Reflect TEACHER KEY

Name _____ Date _____ Period _____

Open **Counting Outcomes Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. How many 2-letter "words" can be made from A, B, C, D if letters can repeat? If they cannot?
   16 with repetition; 12 without.
2. Is choosing a 2-person committee from 4 people the same count as choosing a president and vice-president? Predict both.
   No: committee C(4, 2) = 6, officers P(4, 2) = 12.

2 Explore use the tool

1. Order matters, no repetition: find n and r with 60 outcomes.
   n = 5, r = 3: P(5, 3) = 60.
2. Selections of 2 from 5: group the orderings. How many orderings per selection? How many in all?
   2 orderings each (2!); 10 × 2 = 20 = P(5, 2).
3. Selections from 5: compare r = 2 and r = 3. What do you notice?
   Both give 10: C(5, 2) = C(5, 3) because choosing 2 to take is the same as choosing 3 to leave.
4. 3 scoops from 4 flavors, repeats allowed, order ignored: record the count and two outcomes.
   C(6, 3) = 20; for example AAB, BCD.

3 Record use the tool and fill in the table

Four counting models

n = 5 letters, r = 3 chosen.

Order matters?	Repetition?	Formula	Count
yes	yes	5^3	125
yes	no	P(5, 3) = 5!/2!	60
no	no	C(5, 3) = 5!/(3!·2!)	10
no	yes	C(7, 3)	35

Probability by counting

Situation	Favorable	Total	Probability
Arrange A, B, C, D: starts with A	6	24	1/4
2 letters from A–F with repetition: at least one A	11	36	11/36
Choose 2 of A–E: the pair contains A	4	10	2/5

4 Explain what did you discover?

1. P(n, r) = C(n, r) · r! because each selection can be ordered in r! ways.

2. C(n, r) = C(n, n − r).

3. P(E) = favorable/total only when all outcomes are equally likely.

4. Roll two fair dice. A student counts 21 unordered results and says P(double 6) = 1/21. Explain the error and give the correct probability.

   The 21 unordered results are not equally likely: {1, 2} can happen two ways, {6, 6} only one. Using the 36 equally likely ordered pairs, P(double 6) = 1/36.

5 Apply a new situation, no tool

1. How many 4-digit PINs have no repeated digit?

   P(10, 4) = 5040.

2. A club of 12 chooses a 3-person committee. How many committees?

   C(12, 3) = 220.

3. How many ways can 8 runners finish 1st, 2nd and 3rd?

   P(8, 3) = 336.

4. Flip 5 coins. Find P(exactly 2 heads).

   $C(5, 2)/2^5 = 10/32 = 5/16$.

Teacher notes

Intended discovery: Order and repetition decide the model (n^r, P, C, multisets); each combination corresponds to r! permutations; C(n, r) = C(n, n − r); classical probability needs equally likely outcomes.

Likely misconceptions:

- Using combinations when order matters (officers).
- Treating unordered dice results or multisets as equally likely.
- Adding when the multiplication rule applies.

Answers that can vary: Within the tool's range only n = 5, r = 3 gives 60 in Explore 1. Example outcomes in Explore 4 vary.